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RAPIDLY RISING BREAST CANCER INCIDENCE RATES AMONG ASIAN-AMERICAN WOMEN

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In recent years, breast cancer incidence rates have fluctuated over relatively short time spans; examination of these patterns can provide etiologic clues and direction for prevention programs. Asian-American women are generally considered to be at lower risk of breast cancer than other ethnic groups. However, their rates are typically based on an aggregation of ethnic Asian populations, which may obscure important ethnic differences in risk. Detailed analyses of the trends in ethnic-specific incidence rates will provide more information than when ethnicities are combined. Los Angeles County, California, the most populous and probably the most ethnically diverse county in the United States, has a large multi-ethnic Asian-American population. Trends in invasive female breast cancer incidence were examined using data from the Los Angeles Cancer Surveillance Program, the population-based cancer registry covering the County. Athough overall breast cancer incidence rates remained stable in the late 1980s and early 1990s, data for the most recent 5-year period suggest that incidence may again be increasing for Asian-American and non-Hispanic white women over age 50 (estimated annual percent change = 6.3%, $p < 0.05$ and 1.5%, $p < 0.05$, respectively), although little change has occurred among black and Hispanic women. Invasive breast cancer incidence rates for Asian-American ethnic groups are heterogeneous and, for most, are increasing. In Los Angeles County, rates for Japanese-American women have increased rapidly since 1988 and are now approaching rates for non-Hispanic white women. Rates among Filipinas, who have historically had higher rates than their other Asian-American counterparts, are not increasing as rapidly as rates for Japanese women, but remain relatively high. Breast cancer risk among women of Japanese and Filipino ancestry is twice that of Chinese and Korean women. Asian women, who commonly have low breast cancer rates in their native countries, typically experience increasing breast cancer incidence after immigrating to the United States. Ethnic-specific incidence rates show that Japanese-Americans, the first Asian population to immigrate to Los Angeles County in large numbers and the most acculturated, have experienced a rapid increase in breast cancer incidence. Japanese-American rates in Los Angeles County may have already surpassed those of non-Hispanic whites if recent trends have continued unabated.

Key words: *breast cancer, Asian, Japanese, epidemiology*

Breast cancer incidence rates vary considerably around the world, ranging from a high of more than 70/100,000 women in North America, Israel and temperate South America,[1] to intermediate in much of Europe and Australia, with lows of less than 35/100,000 women in most Asian countries. The differences clearly reflect an unequal distribution of genetic and/or environmental risk factors, from location to location and over time within the same location.

The annual age-adjusted incidence rate of invasive female breast cancer in the United States increased nearly 30% from approximately 85/100,000 women in the mid-1970s to approximately 110/100,000 women in the late 1980s, based on Surveillance, Epidemiology and End Results (SEER) incidence data,[2] yet little overall change was observed for the period 1988 through the mid-1990s. *In situ* breast cancer has followed a roughly similar pattern throughout the late 1980s, but increased throughout the 1990s. Rates for Los Angeles County have been similar to SEER rates. In some populations breast cancer incidence rose during the 1990s when U.S. rates had stabilized.[3,4] For example, breast cancer continues to increase rapidly in Japan and is predicted to become the leading cancer site among native Japanese women in the near future.[1] In some areas, trends may be obscured by divergent patterns according to ethnicity and age.

Ethnicity and national origin are among the strongest predictors of breast cancer risk among women exhibiting at least a 10-fold variation around the world[5]and are associated with relative risk of greater magnitude than those of well-known factors such as ages at menarche, menopause and age at first full-term birth. Historically, non-Hispanic whites and blacks have had higher breast cancer rates than other ethnicities in the United States, whereas rates in Asian-Americans have reflected the lower rates typical of Asian countries. It is well recognized that breast cancer incidence in Asian immigrants shows some shift over several generations toward the rates of non-Hispanic white populations, but it is also generally thought that such populations continue to have low risk relative to other ethnic groups.[6]

Cancer incidence in the large and ethnically diverse population of Los Angeles County, California, is monitored by the Los Angeles Cancer Surveillance Program (CSP). We used data from the CSP and other sources to examine recent trends in the incidence of female invasive breast cancer by ethnicity and age.

MATERIAL AND METHODS

Los Angeles County is the most populous and probably the most ethnically diverse county in the United States, with an estimated 9.7 million residents in 1999. The population-based CSP has collected cancer incidence data for Los Angeles County since 1972. Since 1988, the entire population of California (34.1 million in 1999) has been covered by the population-based California Cancer Registry (CCR), a consortium of regional registries, with the CSP providing Los Angeles County data. The National Cancer Institute's SEER program is also a consortium of state and local population-based registries that has collected cancer incidence data

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since 1973 and has included CSP-collected data from Los Angeles County since 1992. Data from these sources were used in these analyses.

Data from the CSP and other regional registries in California are included in the CCR's Public Use File, 1988–1997. County-level population estimates for the same period were obtained by the CCR from the California Department of Finance (DOF) Demographic Research Unit. In these data, ethnicity is categorized as non-Hispanic white, non-Hispanic black, Hispanic and non-Hispanic Asian/Pacific Islander. Age-adjusted (1970 U.S.) and age-specific incidence rates of female breast cancer in Los Angeles County were calculated by ethnicity using SEER*Stat software.[7] The estimated annual percent change in incidence rates was used to assess the magnitude of changes and linear regression was used to test the significance of trends. Comparisons with rates and trends from the rest of California have been made where appropriate.

Female breast cancer incidence rates in Los Angeles County for all ethnic groups combined were also compared with those of the SEER program, calculated from the SEER Cancer Incidence Public-Use Database, 1973–1997. Because of the large population of Los Angeles County, the CSP contributes annually about 26% of all newly registered cases to the CCR and 21% of all SEER incident cases. Data from Los Angeles County were excluded when comparisons were made with CCR and SEER data.

Due to relatively small numbers of cases and lack of available ethnic-specific population estimates, Asian populations are usually combined for calculation of incidence rates. However, the CSP has prepared annual sex-, age- and ethnic-specific population estimates for Los Angeles County to allow more detailed examination of cancer patterns and trends in individual Asian-American populations, including Chinese, Japanese, Koreans and Filipinos. American Indians, Pacific Islanders, other Southeast Asians and non-specified populations are classified as "other" ethnicity and are not included in the current analyses, due to the small number and heterogeneity of the group.

Ethnicity-specific census population counts for the years 1970, 1980 and 1990 form the basis of the CSP population model for specific Asian subgroups. For the intercensal years of 1972–1979 and 1981–1989, annual population estimates were obtained by linearly interpolating between census years. For 1991–1997, we obtained the annual population estimates for Asian-Americans in Los Angeles County by applying the DOF estimates of age- and sex-specific annual population growth rates to the Asian/Pacific Islander population in Los Angeles County to the 1990 census results. With the CSP population estimates, we calculated age-adjusted invasive breast cancer incidence rates for Chinese, Japanese, Filipino and Korean women. Rates were standardized to the 1970 U.S. population.

RESULTS

Annual invasive breast cancer incidence counts and age-adjusted rates by ethnicity for Los Angeles County are shown in Table I. Over the 5-year period 1993–1997, the breast cancer incidence rate in Los Angeles County rose by an average of 1.1% ($p = 0.11$) per year in non-Hispanic whites, 2.1% ($p = 0.11$) in Hispanics and 4.6% ($p = 0.07$) in Asians, but declined marginally by 0.3% ($p = 0.67$) annually among black women. Comparative data for the CCR excluding Los Angeles County show a somewhat similar pattern over the same 5-year period, with increasing rates in non-Hispanic white (0.9% per year, $p = 0.11$) and Asian women (4.2%, $p = 0.01$). Black women showed a very modest increase (0.2%, $p = 0.85$). However, rates decreased slightly among Hispanic women (−0.1%, $p = 0.88$).

The 1997 SEER rate of 115/100,000 for all ethnic groups combined is the highest annual rate ever observed. SEER data show increases during the 1993–1997 period that are almost identical to Los Angeles County for non-Hispanic white and Asian women, 1.3% ($p = 0.01$) and 4.7% ($p = 0.05$) per year, respectively, with marginal and nonsignificant changes among Hispanic (−0.4% per year, $p = 0.88$) and black women (0.4% per year, $p = 0.25$).

The highest invasive breast cancer incidence rates in Los Angeles are consistently seen among non-Hispanic white women; these rates are about 20% higher than for blacks and roughly double those of Asians and Hispanics (Fig. 1).

For the 5-year period 1993–1997, the greatest increase in invasive breast cancer rates in Los Angeles has occurred among women over 50 years of age (Fig. 2). For non-Hispanic whites, the estimated annual percent increase in age-specific incidence rates for women in this age group was 1.5% ($p < 0.05$) and among

TABLE I – NUMBER OF INCIDENT CASES OF INVASIVE FEMALE BREAST CANCER AND AGE-ADJUSTED INCIDENCE RATES BY ETHNICITY AND YEAR OF DIAGNOSIS, LOS ANGELES COUNTY, CALIFORNIA, 1988–1997[1]

Year	No. of Incident Cases				Rate/100,000			
	NH White	NH Black	Hispanic	NH Asian/PI	NH White	NH Black	Hispanic	NH Asian/PI
1988	3556	470	590	257	134.9	99.8	65.8	66.2
1989	3229	455	654	258	123.7	97.3	67.9	60.1
1990	3222	477	652	304	123.8	99.2	64.8	65.9
1991	3101	487	688	308	118.9	99.5	64.8	62.5
1992	3178	505	761	380	122.3	99.5	67.8	73.3
1993	3139	499	678	362	122.4	98.6	58.1	67.1
1994	3097	530	761	354	122.6	102.8	62.5	62.8
1995	3078	530	760	430	121.6	101.0	59.9	72.8
1996	3080	538	853	459	124.3	100.3	63.8	73.9
1997	3217	546	883	508	128.6	98.4	63.7	77.6
	No. of incident cases				Rate/100,000			
	Chinese	Japanese	Filipino	Korean	Chinese	Japanese	Filipino	Korean
1988	46	59	89	18	42.5	69.5	81.5	26.1
1989	57	48	81	20	50.2	53.9	66.5	26.7
1990	43	66	109	20	34.5	74.7	89.4	24.2
1991	58	52	103	25	43.7	63.5	78.5	28.2
1992	76	106	115	20	51.7	115.7	86.0	21.4
1993	77	75	111	32	54.5	81.2	79.9	31.2
1994	76	70	114	28	49.5	75.3	77.5	29.5
1995	79	93	142	23	47.6	99.8	91.8	21.4
1996	81	91	152	34	47.5	102.5	94.5	32.6
1997	90	98	161	51	50.8	114.3	98.1	44.5

[1]NH, non-Hispanic; PI, Pacific Islander.

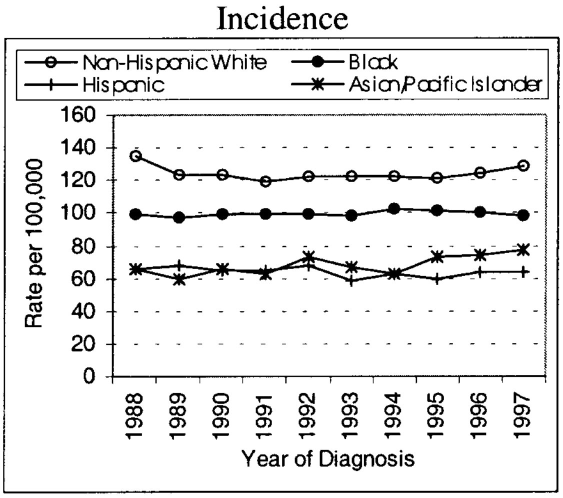

FIGURE 1 – Annual age-adjusted incidence (standardized to the 1970 U.S. population) of invasive female breast cancer by ethnicity, Los Angeles County, California, 1988–1997.

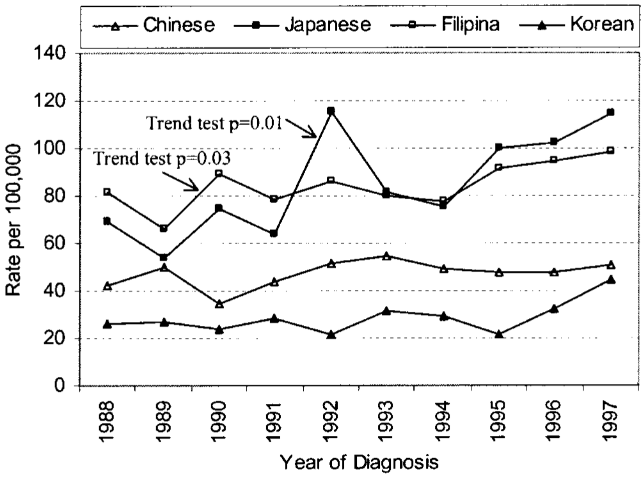

FIGURE 3 – Annual age-adjusted incidence rates (standardized to the 1970 U.S. population) of invasive female breast cancer by Asian ethnic groups, Los Angeles County, California, 1988–1997.

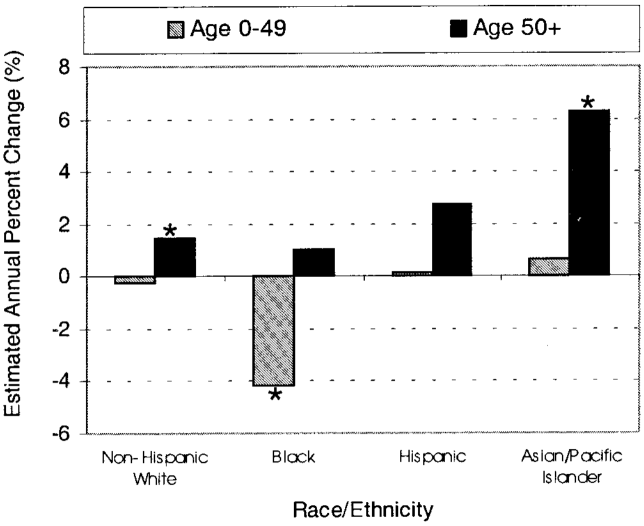

FIGURE 2 – Estimated annual percent change in age-specific incidence rates of invasive female breast cancer (standardized to the 1970 U.S. population) by age at diagnosis, Los Angeles County, California, 1993–1997. *, $p < 0.05$.

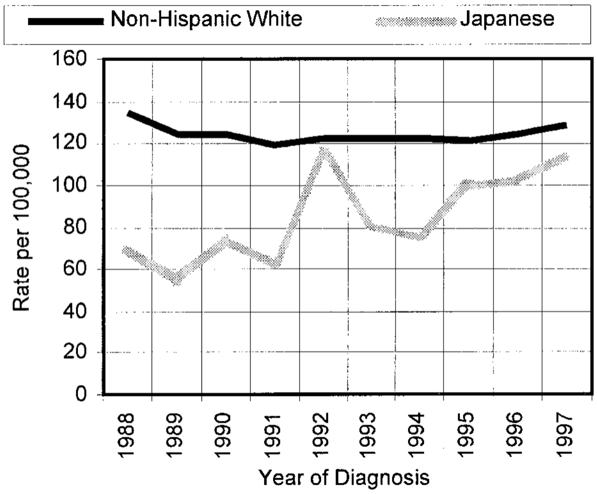

FIGURE 4 – Annual age-adjusted incidence rates (standardized to the U.S. 1970 population) of invasive female breast cancer 1988–1997 for non-Hispanic white and Japanese women in Los Angeles County, California.

Asian women it was 6.3% ($p < .05$); similar increases are seen in CCR and SEER rates for the period (not shown).

The most recent incidence rates for Japanese and Filipino women are approximately double those for Chinese and Korean women and rates for all except Chinese women rose during the 1993–1997 period (Fig. 3). The 1997 rate of 114/100,000 for Japanese-Americans exceeds the rate for black women and is approaching that of non-Hispanic white women in Los Angeles County. Japanese-American invasive breast cancer rates in Los Angeles County may have already surpassed those of non-Hispanic whites, if recent trends have continued unabated (Fig. 4).

DISCUSSION

These analyses document a recent increase in breast cancer incidence, particularly among Asian-American women. Breast cancer incidence for Japanese-American women in Los Angeles County is the highest reported anywhere in the world for Japanese and is nearly as high as the rate for Los Angeles County non-Hispanic whites. In Japan, incidence rates more than doubled from 1960 to the late 1980s[8] and continue to rise.[1] Every birth cohort in Japan from 1890 to 1950 has shown a higher breast cancer incidence at any given age than did the preceding cohort.[9] The Japanese lifestyle has become increasingly Westernized over the past several decades, resulting in adverse trends in breast cancer statistics. Japanese women now marry later, have fewer children, are taller, engage in less physical activity and experience increasing rates of obesity than did previous generations.[1,8] However, changes in the prevalence of 4 major risk factors for breast cancer in Japan (i.e., age at menarche, age at first birth, age at menopause and parity) have been estimated to account for less than 40% of the increase in incidence between 1960 and the 1980s.[8] Extraordinary dietary changes have occurred in urban Asians, particularly the proportion of energy intake derived from fat[10,11] and a decline in consumption of soy products.[12] It is highly plausible that these dietary changes can account for a substantial part of the unexplained increase in incidence over the past several decades.

Several studies have shown that when Asian women migrate to the United States, breast cancer risk increases in subsequent generations.[6] In fact, we have shown through studies in Los Angeles County that there is a substantial increase in the migrating generation itself if the migration occurs quite early in life.[13] Japanese represent the first Asian population to migrate to Los Angeles County in substantial numbers, followed by Filipinos.[14,15] Chinese and Koreans represent less acculturated immigrant populations. It seems likely, based on what has happened among Japanese women in Los Angeles County, that these populations will demonstrate rising breast cancer incidence in the future, as they continue to adopt more Westernized lifestyles.

The plateau in breast cancer incidence rates observed in U.S. populations from the mid-1980s to the mid-1990s was encouraging after many years of increasing rates. *In situ* breast cancer followed a roughly similar pattern through the late 1980s, but incidence rates for *in situ* disease have continued to increase throughout the 1990s.[2] Mammographic screening is no doubt accountable for some of the increase in invasive disease in the previous period.[16–18] However, changing exposures and host factors may also influence incidence. It may be that women born in the post-World War II era, who are only now beginning to reach the age range of high breast cancer incidence, are at greater risk of breast cancer than their mothers, due to such factors as earlier menarche, delayed child-bearing, greater prevalence of obesity, smaller family size and use of hormone replacement therapy.

Although all women are at risk of breast cancer, great variations in risk remain among ethnic groups. Although cancer registries in the United States have long provided incidence rates for blacks and non-Hispanic whites, reliable rates for Hispanics and Asians at the state or regional level are often unavailable. Their small numeric representation in many areas and the lack of annual age-, sex- and ethnic-specific population data to serve as the denominators of rates preclude calculation of reliable rates in most areas. Providing intercensal estimates is particularly difficult for some ethnic populations due to rapidly changing migration patterns.

Los Angeles County offers a favorable environment in which to evaluate cancer incidence patterns. As a center of international immigration and with an ethnically diverse population of nearly 10 million, many ethnic populations are represented in the hundreds of thousands. Although "official" annual population estimates are not available for some of these populations, we use simple techniques to estimate population counts. In some instances, the annual rates we estimate are the only available cancer incidence estimates for certain ethnic groups and provide a means of distinguishing unique patterns among some groups that are commonly aggregated.[19] We show lack of homogeneity in rates across Asian-American populations, with substantial differences in absolute incidence rates and in secular trends in breast cancer among the 4 largest Asian populations in Los Angeles County.

The potential effects of misclassification of ethnicity and errors in population estimates should be considered in interpreting these results. Cancer patient ethnicity information is reported by hospital cancer registrars to the CSP as mandated by California law requiring cancer reporting and is based on information recorded in the medical record. How often this is self-reported *vs.* based on observation or birthplace is unknown. We expect that ethnic misclassification is most likely to classify Japanese as "other Asian," which would result in an underestimation of the true Japanese rate. Because DOF does not provide sex-, age- and Asian subgroup-specific annual population estimates for Los Angeles County for 1991–1997, we applied the growth rate for all Asian/Pacific Islanders to Chinese, Japanese, Filipino and Korean census counts. This may overestimate the Japanese population, since their population increase in previous decades has been lower than the average increase of other Asian/Pacific Island groups. Overestimation of population counts will also produce an underestimate of incidence rates. Thus, the increase in breast cancer incidence among Japanese-American women in Los Angeles County may be slightly greater than the estimates we present.

These data demonstrate the need for increasing awareness among Asian women and their health care providers of breast cancer as a significant health hazard, especially for women of Asian descent. It is possible that many physicians are unaware that the well-known low breast cancer risk among these women in past decades is no longer true and that breast cancer screening is as important as among white and African-American women. The rapid increase in breast cancer incidence rates in Japanese suggests a prominent role of exogenous and, therefore, potentially alterable factors. These data also demonstrate that, in the absence of ethnic-specific cancer incidence rates, important cancer incidence trends may be unrecognized.

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